Constraining the multiplicity of Y dwarfs down to planetary masses at 0.5-1000 au with JWST

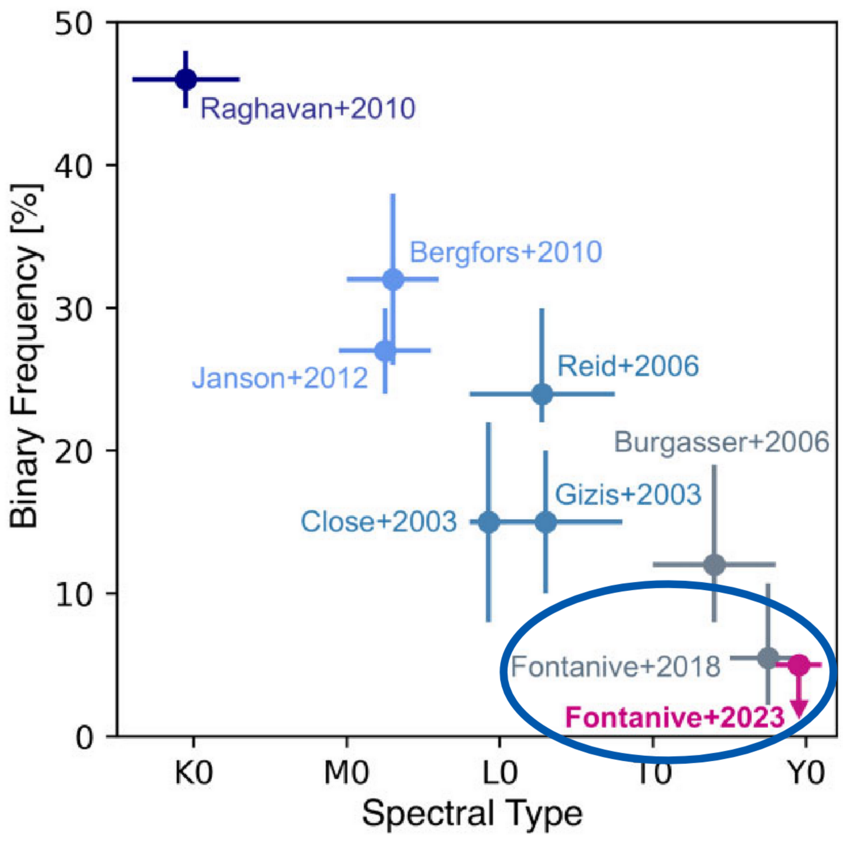
Institut Trottier de recherche sur les exoplanètes
Trottier Institute for Research on Exoplanets
iREx
Université de Montréal

Thomas Vandal [1] C. Fontanive [2,1] M. Giovinazzi [3] L. Albert [1]
R. Doyon [1] F. Martinache [4] D. Mary [4] J. Kammerer [5]
P. Calissendorff [6] M. De Furio [7,6] M. Meyer [6]

[1]**Université de Montréal – Institut Trottier de Recherche sur les exoplanètes (iREx)**
[2]Institute for Astronomy - University of Edinburgh [3]Amherst College [4]Université Côte d'Azur
[5]European Southern Observatory [6]University of Michigan [7]University of Texas at Austin

thomas.vandal@umontreal.ca

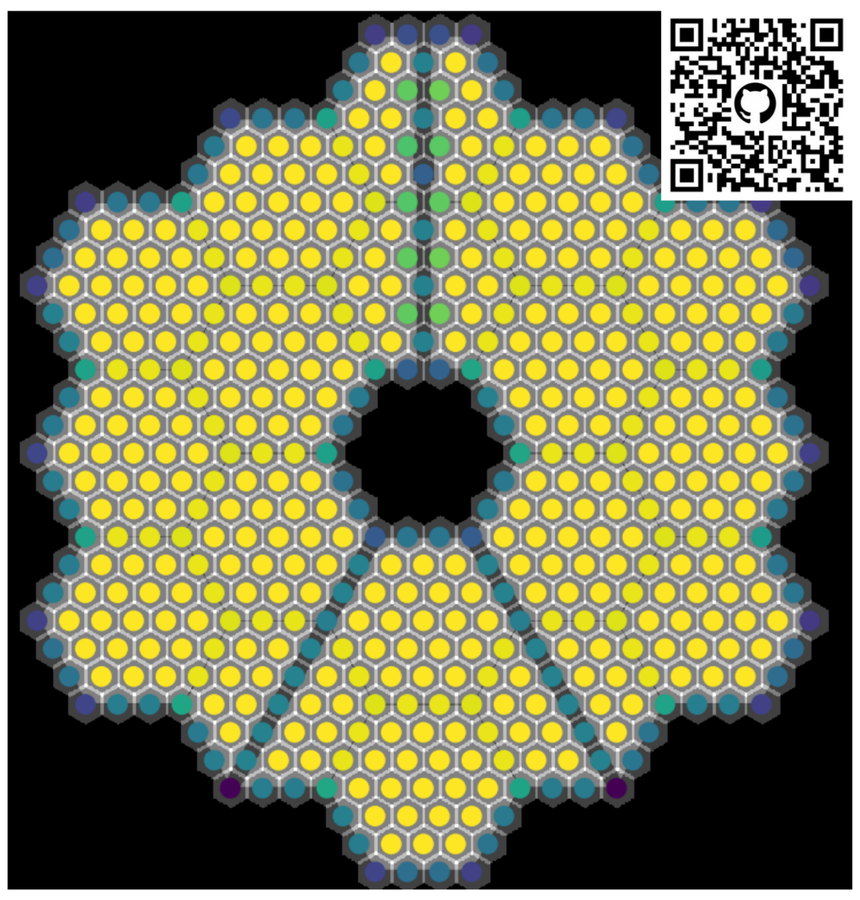
Multiplicity survey of 22 Y dwarfs with JWST



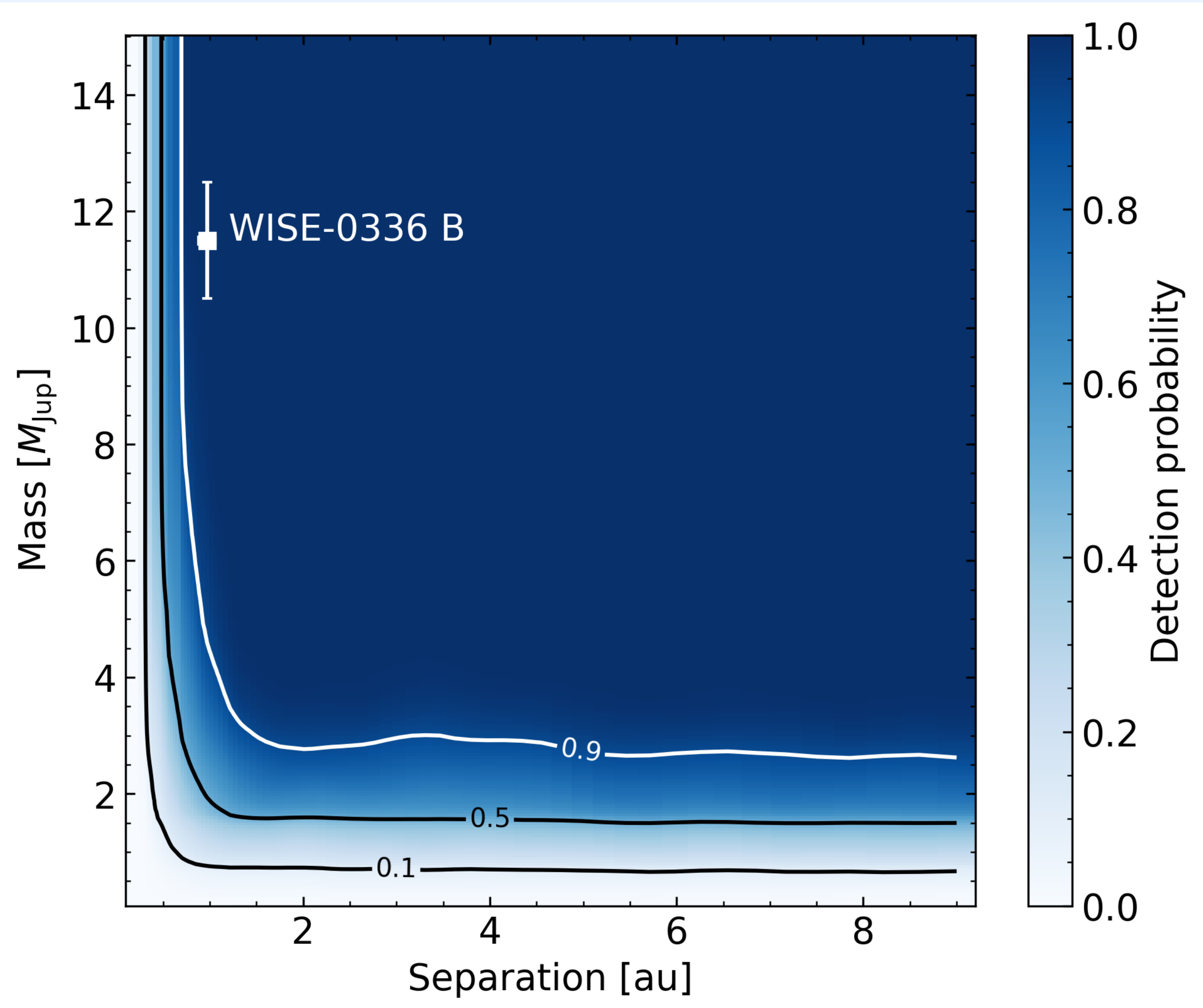
- Multiplicity can be used to distinguish stellar formation mechanisms that produce different binary fractions
- Multiplicity decreases throughout the brown dwarf sequence
- However, for Y dwarfs HST results were limited to a few objects and separations beyond 2 AU [1, 2]
- To confirm this trend and probe short separations, **we surveyed 22 Y dwarfs with JWST NIRCam and NIRISS at 4.8 μm**

Kernel Phase: imaging below the diffraction limit

- We analyze the JWST images with Kernel Phase Imaging: we **model the telescope as an interferometer** made of small sub-apertures [3]
- This unlocks relatively high contrasts below the diffraction limit of JWST
- The pipeline is available on GitHub!
- We **recover the Y+Y dwarf binary WISE-0336 AB** with this technique and obtain parameters consistent with ePSF fitting [4]

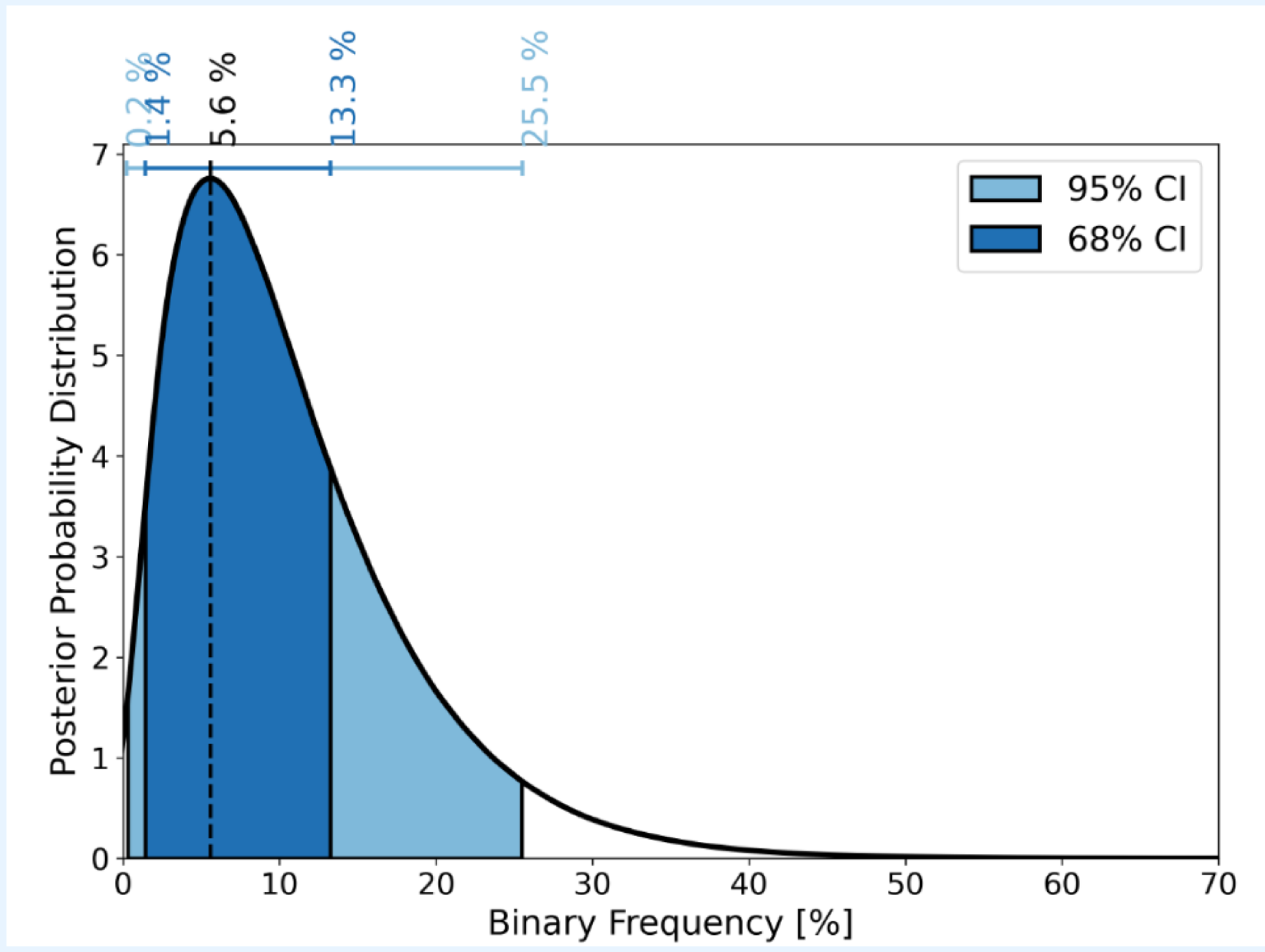


We are sensitive to companions down to 3 M_{Jup} below 2 AU with 90 % completeness and constrain to Y dwarf multiplicity to $5.6^{+7.7}_{-4.2}$ %



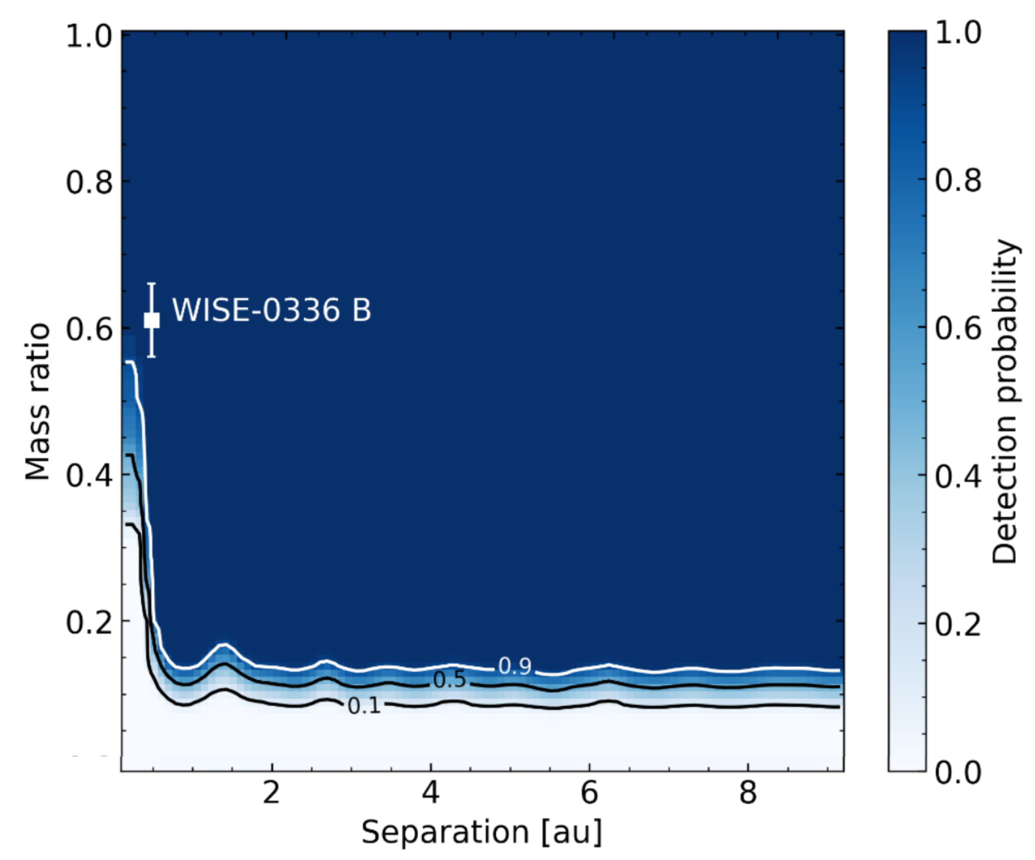
We use our kernel phase contrast limits to derive completeness map for the survey (left)

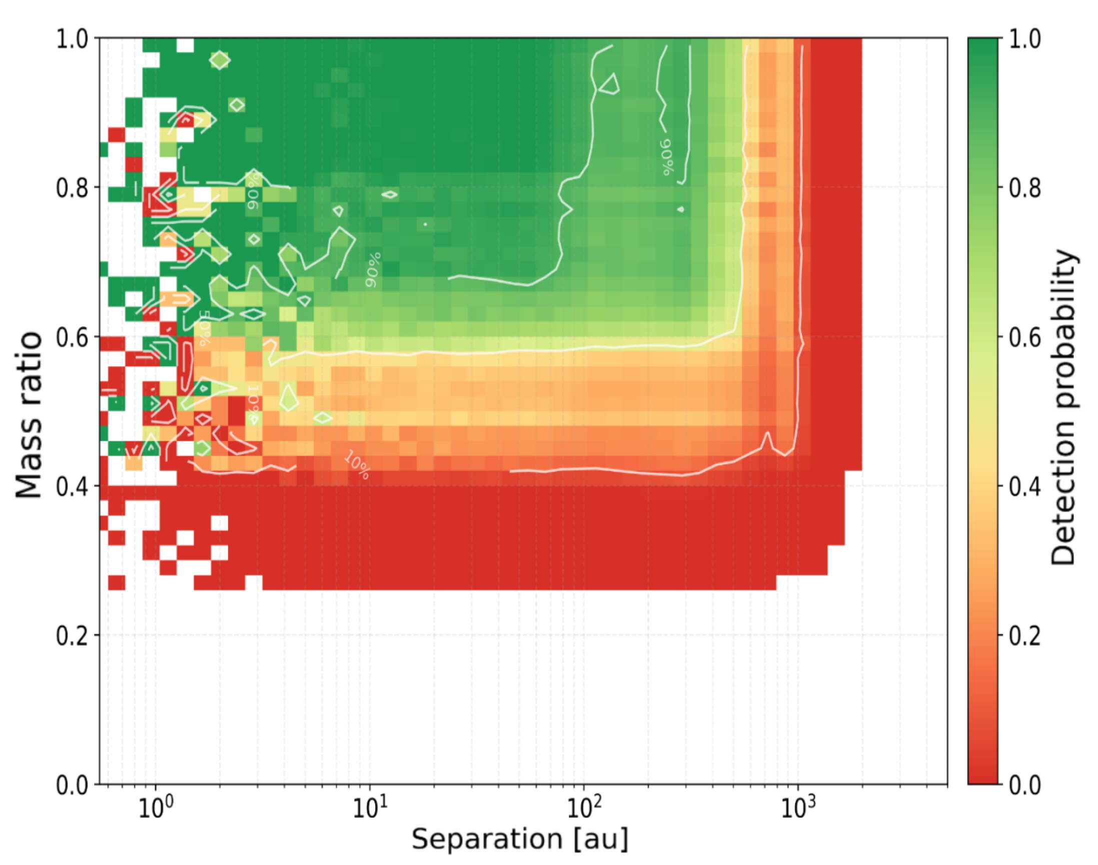
We then use this completeness map and synthetic populations to estimate the binary fraction in a Bayesian framework (right)



We also performed a wide separation search up to 1000 au

No wide companions were found. We are currently combining the completeness maps in the two regimes to derive a global completeness and binary fraction.





Future Work

- Papers coming up: Vandal et al. (in prep), Giovinazzi et al. (in prep)
- Detailed comparison between kernel phase and ePSF fitting
- JWST Cycle 5 Survey 10110 (P.I. Fontanive, Co-P.I. Vandal) to search for companions around more of the coolest brown dwarfs
- **Follow-up WISE-0336**: Track the orbit to get dynamical mass measurements

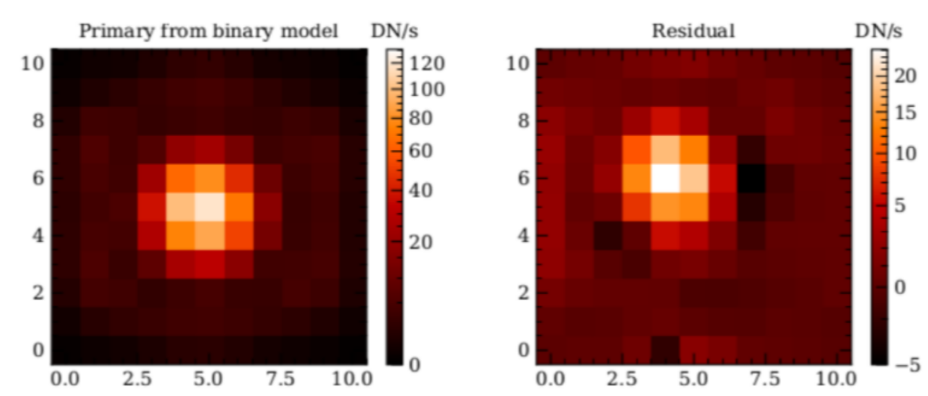


Figure showing the ePSF discovery of WISE-0336 B [4]

References

[1] Fontanive et al., 2018, *MNRAS*, 479, pp. 2702–2727.
[2] Fontanive et al., 2023, *MNRAS*, 526, pp. 1783–1798.
[3] Martinache, 2010, *ApJ*, 724, pp. 464–469.
[4] Calissendorff et al., 2023, *ApJ*, 947, p. L30.

See the PDF version of this poster!

AstroQuébec
Québec
Fonds de recherche – Nature et technologies
Fonds de recherche – Santé
Fonds de recherche – Société et culture